

DDoS Attack Detection in Business Blockchain Networks: A Review, Framework, and Challenges

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Abstract—With the rapid growth of the Internet of Things and the increasing deployment of blockchain technology in critical applications, distributed denial of service (DDoS) attacks pose a significant threat to the security and availability of blockchain networks. However, due to the lack of data sharing among different business chains, existing attack detection systems fail to identify global DDoS attack patterns. This leads to reduced detection accuracy and timeliness, as well as increased risks of false positives and false negatives. Moreover, it prevents the implementation of distributed collaborative defense mechanisms, allowing attackers to bypass local defenses through cross-chain attacks. This paper reviews the current state of DDoS attack detection in blockchain networks and provides a preliminary look at the challenges facing the field. To address these challenges, this paper proposes a new intrusion detection system, BAFL, which utilizes supervision chain and federated learning to detect DDoS attacks in blockchain network layer. We design a distributed DDoS architecture that utilizes a supervised chain of directed acyclic graph (DAG) structures to achieve trusted sharing of model parameters among client nodes and record detected DDoS attacks. The system uses asynchronous federated learning algorithm, weights the client model based on its DDoS attack detection accuracy, and introduces a time attenuation factor to reduce the influence of old model parameters on aggregation and accelerate model convergence. Through comprehensive experiments on data sets such as CIC-DDoS2019, we demonstrated that the accuracy of BAFL in detecting DDoS attacks is more than 99%, which is better than traditional methods, the accuracy is increased by 2%, and the real-time detection delay is reduced by half. The results highlight the robustness and scalability of BAFL, making it a promising solution for protecting blockchain networks from DDoS threats.

Index Terms—DDoS attack detection, federated learning, blockchain, supervision chain.

I. RESEARCH BACKGROUND

THE Internet of Things (IoT) aims to enable smart manufacturing, smart factories and optimized communications within the industrial ecosystem. Due to its ability to share data efficiently and prevent data tampering, blockchain has been widely applied in the IoT. The distributed ledger characteristic of blockchain allows data to be stored and shared across

multiple nodes, avoiding the bottlenecks in data management and processing that occur in centralized architectures, and enhancing the system's scalability and fault tolerance. Moreover, the immutability and transparency of blockchain increase the credibility of data, enabling IoT devices to establish a trust mechanism, ensuring the authenticity and reliability of data, and thus providing a solid foundation for the collaborative operation of IoT devices. However, its interconnected nature and valuable data make it vulnerable to cyber attacks, and without effective protection strategy, the potential cost of cyber attacks will reach \$90 trillion by 2030. Major risks include malware proliferation and downtime caused by DoS and DDoS attacks that exploit system vulnerabilities.

Specifically, the extensive connectivity of IoT devices, which often possess limited computational and storage capabilities and weak security measures, provides attackers with abundant attack vectors and diverse pathways to launch sophisticated and covert DDoS attacks. The substantial data traffic generated by these IoT devices not only facilitates the creation of false traffic surges by attackers, thereby triggering DDoS attacks, but also poses a threat to the normal operation of blockchain networks due to potential data tampering or forgery. The decentralized nature of blockchain networks further complicates the situation, as it allows attackers to bypass traditional defense mechanisms by exploiting vulnerabilities in smart contracts or by controlling multiple nodes, while also dispersing attack traffic across numerous nodes to enhance attack stealth. Moreover, the increasing reliance of critical applications on blockchain networks means that DDoS attacks can not only paralyze these applications, causing significant societal and economic impacts, but also disrupt data sharing mechanisms, affect data synchronization and verification, and potentially infringe upon user privacy. Finally, the diversity of IoT devices and the complexity of data traffic present significant challenges for DDoS attack detection, necessitating the development of novel detection techniques and lightweight defense mechanisms tailored to the IoT environment.

A. DDoS Attacks Detection in Business Blockchain Networks

With the widespread application of blockchain technology in fields such as finance and supply chain management, the risk of DDoS attacks it faces has become increasingly prominent. The decentralized nature of blockchain networks diversifies the attack paths, enabling attackers to launch complex attacks by exploiting consensus mechanisms and smart contracts, such as transaction spamming and Sybil attacks. These attacks not only

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prolong transaction finalization times but also lead to a decrease in network throughput and service performance. Therefore, researching effective DDoS attack detection methods is crucial for ensuring the security and stability of blockchain networks [1].

DDoS attack detection methods in blockchain networks significantly differ from those in industrial settings, mainly in terms of network characteristics, attack patterns, detection technologies, and application scenarios. The decentralized nature, dynamic topology, and complex attack patterns (e.g., transaction spamming and Sybil attacks) of blockchain networks increase detection difficulty. In contrast, industrial networks' hierarchical, fixed topology and relatively singular attack patterns (e.g., network-layer flooding attacks) make detection more focused on device security and mitigation of common attacks. In terms of detection technology, blockchain networks often employ deep learning techniques (such as CNN and LSTM) and decentralized federated learning to accommodate their high dynamism and concurrency. Industrial networks, on the other hand, rely more on traditional machine learning and statistical analysis methods. Additionally, blockchain networks' application scenarios (such as finance and supply chain management) have extremely high requirements for data real-time and security, necessitating rapid identification and mitigation of attacks. Industrial networks, however, place greater emphasis on the stability and reliability of devices. These reasons lead us to the necessity of researching DDoS attack detection methods applicable to blockchain networks.

In addition to localized DDoS attacks targeting individual business chains, the existence of global DDoS attacks within blockchain networks poses a significant threat. These global attacks not only compromise the performance and security of individual business chains but also have the potential to destabilize the entire blockchain network through cross-chain attacks. The interconnected nature of blockchain networks means that an attack on one chain can propagate to others, exploiting shared nodes, protocols, and trust mechanisms to achieve a broader disruption.

The current state of DDoS attack detection in blockchain networks is characterized by a focus on local analysis. Existing detection methods predominantly concentrate on analyzing network traffic data at individual nodes or within specific business chains. While this approach can provide insights into localized attack patterns, it falls short in capturing the full scope of global attacks. The lack of comprehensive, cross-chain data sharing and coordinated analysis mechanisms limits the ability to identify and mitigate attacks that span multiple business chains. This fragmented approach to detection creates vulnerabilities, as attackers can exploit the isolation of individual chains to launch coordinated attacks that evade local defenses and disrupt the overall stability of the blockchain network.

To address these challenges, there is a pressing need for detection frameworks that can integrate data from multiple business chains and perform global analysis. Such frameworks should be capable of identifying attack patterns that transcend individual chains and coordinating defensive measures across the entire network. This requires not only technical advancements in data sharing and analysis but also the development of trust

mechanisms that ensure the integrity and confidentiality of shared information. By adopting a holistic approach to DDoS attack detection, blockchain networks can enhance their resilience and maintain the security and stability necessary for their widespread adoption and operation.

B. Supervision Chain

Blockchain technology effectively addresses the issues of data tampering and information silos prevalent in traditional regulatory models through its immutable and distributed ledger characteristics [2]. It ensures that once data is recorded, it cannot be altered by employing cryptographic techniques and consensus mechanisms. It also supports data sharing and verification across institutions, thereby enhancing the transparency and credibility of regulatory processes. For example, in the financial sector, blockchain can record detailed information of financial transactions in real-time, helping regulatory authorities quickly detect and address violations while protecting investors' interests. Meanwhile, the complete data history recorded by blockchain supports comprehensive traceability, facilitating post-event audits and investigations by regulatory authorities, such as tracing the source of quality issues in supply chain management. Moreover, blockchain supports dynamic node participation and departure, thereby enhancing the system's scalability and maintaining stability in the face of node failures or network partitions. By ensuring real-time access to the latest model parameters and detection results across all nodes, it improves the timeliness of detection and facilitates rapid information sharing. Therefore, blockchain technology can be employed to construct a supervision chain for detecting DDoS attacks within blockchain networks, ensuring data immutability and traceability and guaranteeing data security and integrity.

The "supervision chain" is a core facility built on blockchain technology, used for formulating and enforcing regulatory rules, as well as verifying and auditing cross-chain data [3]. It is usually controlled by regulatory authorities. It has a close relationship with blockchain, inheriting characteristics such as immutability and decentralization. It is a specific application of blockchain technology in the regulatory field. The supervision chain works in tandem with the business chain. The business chain serves specific business scenarios and is responsible for recording and processing business data. The supervision chain, on the other hand, focuses on regulatory functions, ensuring the immutability and traceability of attack data and the parameters of the model. The business chain needs to meet regulatory requirements and connect with the supervision chain to achieve business compliance. The supervision chain is also related to IoT devices. Data generated by IoT devices is processed by the business chain and is also a potential target for DDoS attacks. By regulating the data of the business chain, the supervision chain indirectly ensures the security of IoT device data, guards against DDoS attacks on IoT devices, and ensures that IoT devices operate in a secure environment. In addition, the supervision chain provides a reliable data-sharing and verification platform for DDoS detection. Through technologies such as smart contracts, it enables the automatic verification and sharing of attack data

that has been detected, thereby enhancing the accuracy and timeliness of DDoS detection and providing strong support for the implementation of defense measures.

C. Introduction to This Work

In the current blockchain and federated learning (FL) integrated architecture, the absence of real-time and reliable data-sharing mechanisms between business chains restricts detection systems to local traffic monitoring. This makes identifying global DDoS attack patterns difficult, reducing detection accuracy and timeliness, and increasing false positives and negatives. Data isolation between chains also hinders distributed collaborative defense, as attackers can exploit information silos for cross-chain attacks. The distributed nature of blockchain networks diversifies attack paths, allowing attackers to exploit consensus mechanism and smart contract vulnerabilities for complex attacks that bypass traditional defenses. This further complicates global attack pattern identification. Additionally, conducting business processes and security monitoring on the same blockchain limits system flexibility and results in suboptimal security performance. The significant differences in data formats and structures between IoT business chains cause inconsistent feature extraction, affecting detection model accuracy and generalization ability.

To address these issues, this paper proposes combining DAG-structured blockchain with federated learning for DDoS attack detection in IoT business chain networks. DAG-structured blockchains offer high throughput and low latency, enabling rapid recording of model parameters and attack events and real-time sharing of model parameters and attack information. Federated learning can integrate heterogeneous network traffic information from different business blockchain networks for distributed detection while protecting the privacy of local network traffic data. By using dynamic weight allocation and time decay mechanisms, federated learning can improve detection accuracy and model convergence speed. Therefore, this combination can effectively address the challenges of DDoS attack detection in blockchain networks and enhance the ability of blockchain networks to resist DDoS threats.

The main contributions of this study are summarized as follows:

- 1) We have conducted a comprehensive and in-depth investigation of DDoS attack detection methods in blockchain networks, systematically summarized the widely adopted technical means at present, and thoroughly analyzed the key challenges and urgent problems currently faced in this research area.
- 2) We propose a cross-chain supervision method called BAFL, which is based on the DAG (Directed Acyclic Graph) structure and employs a federated learning framework. This method analyzes heterogeneous network traffic information from different business blockchain networks and strictly protects the privacy of local network traffic data for each client node and the business chains they supervise during the model training process. It also utilizes smart contract technology to achieve automated data verification and sharing within the supervision chain

based on the DAG structure. Given that DAG chains offer higher transaction throughput compared to traditional blockchains, our method can reduce the latency of on-chain data sharing and enable real-time sharing of model parameters and attack cases among each client node.

- 3) We introduce a time decay mechanism for asynchronous aggregation. When client nodes in federated learning unexpectedly go offline due to crashes, this mechanism reduces the impact of outdated model parameters on the aggregation results, ensuring that the model training remains unaffected and thereby enhancing the scalability of federated learning. Moreover, by leveraging blockchain technology to record the update history of model parameters, we ensure the immutability and traceability of data. When new clients join the federated learning training, they can quickly synchronize with the latest model parameters via the blockchain and thus participate in normal model training.
- 4) To enhance the accuracy of the federated learning model in DDoS detection, we have adopted a dynamic weight assignment method. By assigning greater weights to nodes with higher accuracy in each training round, we allow models with better performance to have a larger proportion in the aggregation process. This approach not only improves the accuracy of DDoS detection but also reduces the number of rounds and time required for model convergence.

The rest of this paper is organized as follows: Section II presents DDoS detection methods in blockchain networks. Section III introduces the Challenges for DDoS attack detection in business blockchain networks. Section IV introduces the types of DDoS attacks in blockchain networks. Section V describes the system model and the details the scheme proposed in this paper. In Section VI, we present experimental simulations. Finally, we conclude in Section VII. The structure of this paper is illustrated in Fig. 1, and a detailed comparative analysis of the related works is presented in Table I.

II. DDoS MITIGATION AND DETECTION METHODS IN BLOCKCHAIN NETWORKS

A. Blockchain Technology-Based Mitigation Methods

Blockchain technology, with its characteristics of decentralization, immutability, and transparency, has demonstrated a certain ability to mitigate Distributed Denial of Service (DDoS) attacks. Through the recording and verification mechanisms of the distributed ledger, blockchain can effectively identify and block the propagation of malicious traffic, thereby reducing the occupation and destruction of network resources by attacks. Moreover, the consensus mechanism of blockchain can ensure that nodes in the network reach a consistent identification and response to attack behaviors, further enhancing the network's defensive capabilities [4], [5]. For example, several blockchain-based mitigation methods have been proposed, including decentralized Content Delivery Network (CDN) schemes [6], smart contract-based attack information sharing [7], and blockchain-SDN collaborative frameworks [8]. These methods leverage

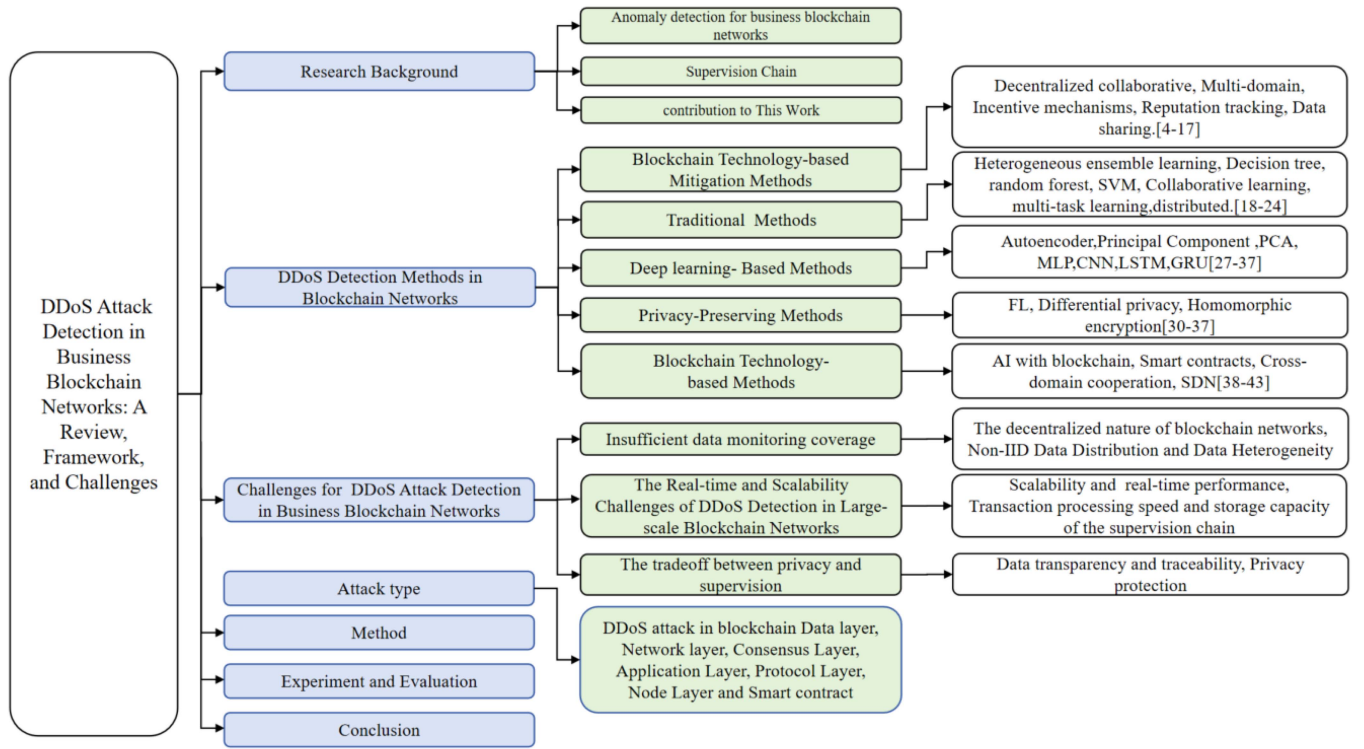


Fig. 1. The structure of this paper.

blockchain's inherent security features to enhance DDoS attack detection and mitigation. [9] systematically studied the attack principles and defense strategies of blockchain from two dimensions—blockchain hierarchical structure and attack association analysis—summarized blockchain attack clusters, and constructed a relatively comprehensive distributed blockchain security defense system. Several scholars have proposed DDoS attack detection and mitigation schemes based on blockchain and SDN. Co-IoT [10], on the other hand, utilizes SDN controllers to detect DDoS attacks in IoT devices and shares attack information via Ethereum smart contracts to block malicious traffic. The BloSS model is a collaborative threat information-sharing framework that integrates blockchain, smart contracts, and software-defined networking (SDN). It utilizes IPFS to store data and ensures data security through the use of blockchain-based hash values [11], [12], [13], [14], [15]. Additionally, the DefenseChain framework employs smart contracts to detect and mitigate DDoS attacks, supporting dynamic strategy updates and traffic redirection [16]. Other studies have explored privacy-preserving mechanisms based on Bloom filters, systems that encrypt IP lists using AES algorithms, and Ethereum-based frameworks for verifying the integrity of IoT sensor data [17].

Although blockchain technology shows great potential in mitigating DDoS attacks, it still faces many challenges in the detection and defense of DDoS attacks, which limit its effectiveness. Therefore, it is necessary to further enhance the DDoS attack detection capabilities within blockchain networks. For example, to maintain the consistency of model parameters across different nodes, it is necessary to overcome the synchronization delay that may be caused by the blockchain consensus mechanism, which

directly affects the efficiency of model updates. Moreover, after detecting an attack, the system needs to take defensive measures quickly, but communication latency and node response time in distributed systems may hinder the timely implementation of defensive measures.

B. Traditional Detection Methods

Traditional DDoS attack detection methods in blockchain networks mainly include machine learning-based techniques and defense mechanisms that leverage blockchain characteristics. Traditional methods include machine learning-based anomaly detection [18], fee-based mempool optimization [19], and collaborative anomaly detection frameworks [20]. These methods leverage various algorithms and optimization strategies to enhance detection accuracy and efficiency. [21] analyzes the behavior of the Ethereum network under low-cost DDoS attacks and proposes using machine learning techniques, such as decision trees and random forests, to detect such attacks. These approach can effectively identify DDoS attacks and other abnormal behaviors, and further determine the type of attack through clustering analysis. Some research focus on the issue of DDoS attack detection in Blockchain-IoT systems and propose distributed detection frameworks based on fog computing and machine learning. [22] presents a distributed framework that integrates IPFS and optimized feature selection, employing Random Forest (RF) and XGBoost algorithms, and is suitable for smart contract-based Blockchain-IoT systems. [23] is based on heterogeneous ensemble learning (AdaBoost and Random Forest) and virtual-reality parallel blockchain technology,

TABLE I
A COMPARISON OF RELATED WORKS

Scope	Ref.	Year	Emphasis	Summary
DDoS Detection Methods in blockchain networks	[1]	2021	Survey	This paper reviews various methods for mitigating Distributed Denial-of-Service (DDoS) attacks using blockchain technology across different fields to date.
Supervision Chain	[2]	2023	cross-chain supervision	This paper designs SCOPE, a cross-chain supervision scheme for consortium chains, using a relay chain for automatic supervision and a method for verifying cross-chain data authenticity via oracle node reputation and weighted threshold signatures.
	[3]	2022	cross-chain verification	This paper proposes Surveillant, a blockchain supervision mechanism using dual-functional nodes for efficient cross-chain validation, Merkle Mountain Range for data aggregation, and an improved incentive mechanism.
Blockchain Technology-based DDoS Mitigation Methods	[4]	2020	Blockchain, smart contracts, hash encryption, DDoS defense.	A blockchain-based DDoS defense architecture is proposed, which protects IoT devices from DDoS attacks through smart contracts and hash encryption technologies. The architecture leverages the immutability and distributed nature of blockchain to ensure data security and prevent unauthorized access.
	[5]	2020	Miner incentives, game theory	This paper investigates a novel type of Blockchain Denial-of-Service (BDoS) attack, which disrupts the normal operation of blockchain by manipulating miner incentives. Through game-theoretic analysis, the paper proposes the conditions for the attack and possible mitigation measures.
	[6]	2018	Private blockchain, trusted nodes, decentralized DDoS mitigation	This paper proposes a distributed Content Delivery Network (CDN) schema using private blockchain to address the issue of unreliable nodes in existing centralized CDN systems and provides a DDoS mitigation solution suitable for military and government agencies.
	[9]	2023	Blockchain hierarchical architecture, attack correlation analysis, decentralized federated learning, defense strategies	This research investigates blockchain attacks from two dimensions: blockchain hierarchical architecture and attack correlation analysis. It analyzes the characteristics of DDoS attacks in blockchain networks and proposes defense strategies based on the unique features of blockchain. For example, it suggests using decentralized federated learning to reduce communication overhead and improve detection efficiency.
	[7]	2019	DDoS Mitigation, in-domain mitigation, cross-domain mitigation, smart contracts	The Cochain-SC scheme is proposed. It integrates an in-domain DDoS attack detection and mitigation method based on SDN (including entropy-based detection and Bayesian classifier) with a cross-domain DDoS attack collaborative mitigation mechanism based on blockchain and smart contracts. This approach achieves efficient mitigation along the attack path and effective mitigation near the attack source.
	[10]	2019	Decentralized collaborative mitigation of DDoS attacks, smart contracts	A collaborative mitigation scheme for DDoS attacks in the Internet of Things (IoT) environment based on blockchain and SDN, called Co-IoT, is proposed. It leverages smart contracts to achieve secure, efficient, and decentralized sharing of attack information, thereby enabling collaborative detection and mitigation of DDoS attacks across multiple SDN domains.
	[8]	2017	Multi-domain mitigation, Software-Defined Networking (SDN), Network Function Virtualization (NFV)	A multi-domain DDoS attack mitigation architecture that integrates blockchain, Software-Defined Networking (SDN), and Network Function Virtualization (NFV) is proposed. It leverages the distributed ledger characteristics of blockchain to achieve sharing and dissemination of attack information, while dynamically adjusting network defense strategies through SDN and NFV.

optimizing detection performance for DDoS attack detection in blockchain networks. [24] designs a fog computing-based distributed intrusion detection system, using RF and XGBoost algorithms for distributed detection, emphasizing low latency and efficient detection, and is suitable for Blockchain-IoT

networks. Although these traditional methods can detect and mitigate DDoS attacks to some extent, they usually rely on specific network features or the performance of machine learning models, and may need further optimization and improvement when facing complex attack scenarios.

TABLE I
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Scope	Ref.	Year	Emphasis	Summary
	[11]	2020	Smart contracts, collaborative defense, reputation mechanisms, cross-domain cooperation	This paper introduces the design, implementation, and evaluation of the BloSS system. The system leverages blockchain and smart contract technologies in combination with Software-Defined Networking (SDN) to achieve collaborative detection and mitigation of cross-domain DDoS attacks. It also promotes cooperation among service providers through reputation and incentive mechanisms.
	[12]	2019	Collaborative defense against DDoS attacks, incentive mechanisms, distributed systems	A blockchain-based collaborative defense system against Distributed Denial of Service (DDoS) attacks, called BloSS, is proposed. It leverages blockchain technology to achieve distributed sharing of attack information and to implement incentive mechanisms, enabling different Autonomous Systems (AS) to collaborate in the detection and mitigation of DDoS attacks.
	[13]	2020	Smart contracts, distributed defense, incentive mechanisms, reputation tracking	This paper introduces SC-FLARE, a smart contract-based distributed DDoS signaling protocol deployed on the Ethereum Proof-of-Authority blockchain. It is used for sharing attack information, incentive exchanges, and reputation tracking, in order to achieve fully distributed and automated collaborative defense.
	[14]	2019	DDoS cooperative defense, incentive mechanism, smart contract, reputation system	This paper designs and implements a reputation mechanism for the Blockchain-based Signaling System (BloSS). It automates reputation management through smart contracts and utilizes the Beta Reputation System to identify and reward honest participants, thereby incentivizing cooperative behavior in DDoS defense.
	[15]	2019	Ethereum blockchain, collaborative defense, DDoS attacks, security visualization, decision support	This paper designs a security management system dashboard for the blockchain-based collaborative DDoS defense system BloSS. It enables cybersecurity analysts to monitor attack information in real time and make decisions on whether to request or accept collaborative defense on an interactive interface.
	[16]	2022	Threat intelligence sharing, distributed trust, economic model, reputation system	This paper proposes the DefenseChain platform, which leverages a consortium blockchain architecture to achieve threat intelligence sharing and defense. It evaluates participants based on the indicators of “detection quality” and “mitigation quality,” and incentivizes cooperation through an economic model, effectively identifying rational/irrational service providers.
	[17]	2020	Smart contracts, data confidentiality, data integrity, IP sharing	This paper designs a DDoS defense platform based on a consortium blockchain. It automatically compares suspicious IP addresses uploaded by different Security Operations Centers (SOCs) through smart contracts and ensures data confidentiality and integrity using elliptic curve cryptography, thereby achieving the sharing of malicious IP addresses and attack prevention.
Traditional Methods	[18]	2019	One-Class Support Vector Machine (OCSVM), K-Means clustering, double-spending attack, 51% attack	machine learning-based anomaly detection model is proposed, which uses One-Class Support Vector Machine (OCSVM) to detect anomalous transactions and classifies them using the K-Means clustering algorithm. The model validates its effectiveness by simulating data for DDoS attacks, double-spending attacks, and 51% attacks.
	[21]	2021	Ethereum, machine learning, decision tree, random forest.	The behavior of the Ethereum network under low-cost DDoS attacks is analyzed, and the use of machine learning techniques (such as decision trees, random forests, etc.) to detect such attacks is proposed.

These methods have achieved certain effects in early attack detection. However, in the complex blockchain network environment, their accuracy and real-time performance are limited. First, attackers can exploit the blockchain consensus mechanism to transmit redundant data, and hide DDoS attack traffic in redundant P2P traffic through protocol obfuscation or mutation flooding mechanisms, reducing the accuracy of statistical

detection methods. Second, the consensus mechanism in blockchain processes the traffic load information sent by all miner nodes. Attackers can encapsulate and transmit rumor-based DDoS attack traffic containing disguised load information [5], further reducing the accuracy of statistical detection methods [20]. Moreover, these methods may face performance bottlenecks when dealing with high-dimensional data. On the

TABLE I
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Scope	Ref.	Year	Emphasis	Summary
	[19]	2018	Mempool optimization, fee limit	A method for defending against DDoS attacks by optimizing the Bitcoin mempool is proposed. This method includes fee-based and age-based designs, filtering out spam transactions by limiting transaction fees and transaction age.
	[20]	2018	Collaborative learning, multi-task learning, condition management	A blockchain-based collaborative anomaly detection framework is proposed, which transforms the anomaly detection problem into a multi-task probabilistic dictionary learning problem. Through the use of smart contracts and statistical machine learning algorithms, condition management of industrial assets on the blockchain is realized.
	[22]	2021	Fog computing, Random Forest, XGBoost, IPFS, feature selection	A distributed framework based on fog computing is proposed for detecting DDoS attacks in smart contract blockchain Internet of Things (IoT) systems. The framework employs two machine learning algorithms, Random Forest (RF) and XGBoost, for decision-making and integrates the IPFS distributed file storage technology to achieve data load balancing and decentralized storage. In addition, a feature selection method based on mutual information is designed to reduce data dimensionality and improve detection efficiency.
	[23]	2020	Heterogeneous ensemble learning, AdaBoost, Random Forest	A lightweight DDoS attack detection scheme based on heterogeneous ensemble learning is proposed, which combines the AdaBoost and Random Forest algorithms. The scheme deploys different types of lightweight classifiers (such as CART and ID3) in the blockchain network and leverages virtual reality parallel blockchain technology to optimize detection performance. This method performs well in artificial blockchain networks and can effectively guide the defense against DDoS attacks in real blockchain networks.
	[24]	2022	Fog computing, Random Forest, XGBoost, distributed detection	A distributed intrusion detection system based on fog computing is proposed for detecting DDoS attacks in blockchain Internet of Things (IoT) networks. The system employs Random Forest (RF) and XGBoost algorithms for distributed detection, which is suitable for detecting DDoS attacks in blockchain IoT networks, emphasizing distributed detection and low latency.
Deep learning-Based Methods	[27]	2019	Bitcoin ecosystem, Principal Component Analysis (PCA), Multilayer Perceptron (MLP), feature extraction	The study collects data from the Bitcoin network and DDoS attacks, employs Principal Component Analysis (PCA) for feature extraction, and utilizes a Multilayer Perceptron (MLP) neural network for DDoS attack detection. The experimental results indicate that this method can effectively correlate Bitcoin network data with service-level DDoS attacks.
	[28]	2019	Ethereum smart contracts, RNN-LSTM, collaborative DDoS mitigation, attack information sharing	The study proposes a collaborative DDoS mitigation scheme based on Ethereum smart contracts and Recurrent Neural Network with Long Short-Term Memory (RNN-LSTM). It leverages deep learning to detect DDoS attacks and uses smart contracts to achieve rapid sharing of attack information and collaborative defense.
	[29]	2022	Cross-Multi-Layer Convolutional Neural Network (CMCNN), L2 regularization, Stacked Sparse Autoencoder (SSAE)	The study proposes a DDoS attack detection method based on the Cross-Multi-Layer Convolutional Neural Network (CMCNN). Targeting the attack traffic at the blockchain network layer, the method optimizes model parameters through L2 regularization, Stacked Sparse Autoencoder (SSAE), and an improved stochastic gradient descent algorithm, effectively enhancing the detection performance of DDoS attack traffic.

one hand, the blockchain network layer has various consensus mechanisms (such as PoW, PoS, DPoS, Byzantine fault tolerance, etc.), and the diversity of underlying traffic protocols (such as RLPx, GHOST, Gossip, DEVp2p, Strata, Gnutella, etc.) increases the dimensionality of non-linear features of traffic data and the complexity of classification output of DDoS detection models. This weakens the model's generalization ability for DDoS traffic. What's more, the characteristics of rumor-based

attack traffic in the blockchain network layer are very similar to those of normal network traffic, making it difficult to distinguish the core features of mixed DDoS attack traffic through machine learning methods.

C. Deep Learning-Based Detection Methods

Deep learning demonstrates significant advantages in detecting DDoS attacks due to its powerful capability of extracting

TABLE I
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Scope	Ref.	Year	Emphasis	Summary
Privacy-Preserving DDoS Attack Detection Methods for Blockchain Networks	[30]	2021	Blockchain, Long Short-Term Memory (LSTM) network	This paper proposes a privacy-preserving framework based on blockchain and deep learning, called P2SF-IoV, for secure communication and intrusion detection in the Internet of Vehicles (IoV). The framework achieves secure transmission of IoV data through the blockchain module and utilizes the deep learning module (specifically the Long Short-Term Memory (LSTM) network) to analyze the data provided by the blockchain module, thereby detecting intrusion activities.
	[33]	2021	Long Short-Term Memory Autoencoder (LSTM-AE), Federated Gated Recurrent Unit Neural Network Algorithm (FedGRU)	This paper proposes a federated learning framework based on deep privacy encoding, called PEFL, for privacy protection and intrusion detection in smart agriculture. The framework protects data privacy through perturbation encoding and Long Short-Term Memory Autoencoder (LSTM-AE) technology, and designs a Federated Gated Recurrent Unit Neural Network Algorithm (FedGRU) for intrusion detection.
	[34]	2022	Deep Variational Autoencoder (DVAE), Attention-based Deep Gated Recurrent Neural Network (A-DGRNN)	This paper proposes a privacy-preserving threat intelligence framework based on blockchain and deep learning, called P2TIF, for data protection and threat detection in Industrial Internet of Things (IIoT) environments. The framework prevents data poisoning attacks through the blockchain module and protects data privacy using Deep Variational Autoencoder (DVAE) technology. It then performs threat detection using an Attention-based Deep Gated Recurrent Neural Network (A-DGRNN).
	[35]	2021	Federated Deep Learning, IDS,review	This paper reviews the applications of federated deep learning in the Internet of Things (IoT), especially in the context of privacy-preserving and intrusion detection systems for Industrial Internet of Things, edge computing, drone IoT, healthcare IoT, and vehicular IoT. It also compares the performance of centralized and federated deep learning (including RNN, CNN, and DNN) in detecting attacks through experimental analysis.
	[36]	2021	Blockchain, federated learning, differential privacy, homomorphic encryption	This paper proposes a blockchain-based federated learning data protection aggregation scheme, which combines differential privacy and homomorphic encryption technologies. It is used for data sharing and model sharing in the Industrial Internet of Things (IIoT) to address security vulnerabilities in distributed learning. Experiments have verified its advantages in terms of privacy protection and performance.
	[37]	2024	Federated learning, blockchain, centralized and decentralized, LSTM	This paper proposes a DDoS attack detection architecture based on federated learning, employing both centralized and decentralized federated learning methods combined with Long Short-Term Memory (LSTM) networks. The model is trained and validated using datasets generated by simulating DDoS attacks on blockchain, effectively enhancing the detection performance of DDoS attacks.
	[31]	2022	Federated learning,IDS, Gated Recurrent Unit (GRU).	This paper proposes a federated learning-based anomaly detection method for IoT security attacks. It utilizes the Gated Recurrent Unit (GRU) model for distributed training, sharing only model weights on local devices. The global model is optimized through ensemble learning, effectively protecting user data privacy and improving attack detection accuracy.

high-dimensional features. It can automatically identify subtle patterns and features associated with attacks from complex network traffic data, thereby enabling more accurate attack detection and classification [25], [26]. [27]proposes a deep-learning-based method for detecting DDoS attacks in bitcoin ecosystem by collecting blocks and transaction data from the

Bitcoin network, extracting statistical features, reducing feature dimensions using Principal Component Analysis (PCA), and ultimately predicting and detecting DDoS attacks with a Multi-Layer Perceptron (MLP) neural network.A blockchain-based DDoS mitigation method is proposed in [28], which reduces signaling complexity and enables multiple autonomous systems

TABLE I
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Scope	Ref.	Year	Emphasis	Summary
	[32]	2023	Blockchain, federated learning, differential privacy, Proof of learning	This paper proposes a blockchain-based privacy-preserving security framework for federated deep learning, called PPSS. It employs differential privacy training and a Proof-of-Federated Deep-Learning (PoFDL) consensus protocol, combined with Convolutional Neural Networks (CNN) for intrusion detection, effectively addressing data privacy and security issues in the Industrial Internet of Things (IIoT).
DDoS Attack Detection and Mitigation Method based on Blockchain Technology	[38]	2020	Integrity verification, decentralized attack detection	A blockchain-based security analysis method for Internet of Things (IoT) networks is proposed to prevent malicious attacks and intrusions. This method verifies the integrity of data generated by IoT devices using blockchain technology and leverages the decentralized nature of blockchain to enhance security.
	[39]	2021	Traceroute, distributed detection, RNN prediction	This paper proposes a blockchain-based Link-Flooding Attack (LFA) detection scheme, which deploys blockchain technology on routers and servers to record and share Traceroute information, and uses an RNN model to predict traffic paths, thereby achieving timely detection of LFA.
	[40]	2020	On-chain data storage, Random Forest, feature selection	A DDoS attack mitigation method combining blockchain and machine learning technologies is proposed. This method uses machine learning algorithms (such as KNN, decision trees, and Random Forest) to detect malicious traffic and stores the blacklist on the blockchain to achieve tamper-proof records.
	[41]	2020	Smart contracts, access control, LSTM neural network	A blockchain-based defense method against DDoS attacks on IoT devices is proposed. This method detects abnormal behavior of terminal devices by analyzing network traffic features at edge nodes, and implements DDoS attack defense by deploying information about attack nodes and access control policies through smart contracts in the blockchain.
	[42]	2021	Deep learning, blockchain, zero-knowledge proof, voting consensus	A blockchain framework based on deep learning is proposed to protect software-defined industrial networks. The framework registers and authenticates switches through a zero-knowledge proof mechanism, and adds switches to the blockchain using a voting consensus mechanism, while employing Deep Boltzmann Machines (DBM) to detect anomalous traffic.
	[43]	2020	Smart contract-based DDoS attack detection, fuzzy neural network, privacy protection, data sharing	This paper proposes a blockchain-based DDoS detection system that utilizes a consortium blockchain structure to place participants within a private chain. It detects anomalous data through a fuzzy neural network algorithm embedded in smart contracts and uploads the abnormal results to a public chain to achieve privacy protection and data sharing.
	[44]	2024	SDN, deep learning, moving target defense	This paper presents a blockchain-supported Software-Defined Networking (SDN) framework, BSDN-HMTD, which integrates deep learning methods for DDoS attack detection. It enhances network security and robustness through user authentication, traffic validation, and dynamic moving target defense (MTD) techniques.
	[45]	2023	Smulti-model integration, dynamic weight adjustment	This paper proposes an adaptive DDoS detection and classification method based on blockchain. By integrating multiple deep learning models (GRU, CNN, LSTM, DNN, and SVM) and introducing a dynamic weight adjustment mechanism, the method improves the detection and classification performance for DDoS attacks.

to share hardware resources and defense capabilities. It also integrates a deep learning DDoS detection system to identify and block only the specific malicious traffic, instead of blocking all traffic to the victim. In addition, in the field of AI detection of DDoS attacks, current research and practice tend to adopt centralized training and detection strategies. A DDoS attack detection method based on a cross-multi-layer convolutional

neural network is proposed in [29], targeting DDoS attacks in the blockchain network layer. The method extracts robust features of attack traffic through cross-layer convolution operations with L2 regularization. It employs a stacked sparse autoencoder based on Kullback-Leibler divergence for feature encoding to reduce redundancy and prevent overfitting. Moreover, it optimizes model parameters using an improved stochastic gradient

descent (ISGD) algorithm to accelerate convergence. However, these centralized approach has several limitations in the context of blockchain networks.

Specifically, the limited computing resources of centralized detection organizations make it difficult to process large-scale network traffic data quickly and efficiently, which can lead to insufficient response speed to attacks. A centralized approach can also raise the risk of privacy breaches during the transmission of network traffic, as a large amount of data needs to be transferred across the network, increasing the possibility of data interception and abuse.

D. Privacy-Preserving Detection Methods

At present, in the field of network attack detection, the research of privacy protection methods has made remarkable progress. With the rapid development of network technology and the increasing awareness of data privacy, how to protect user privacy while detecting network attacks has become an urgent problem to be solved. In recent years, researchers have proposed a variety of privacy protection methods to address this challenge. Specific methods include vehicular network security frameworks combining blockchain and deep learning [30], federated learning-based anomaly detection [31], and blockchain-enabled federated deep learning frameworks [32]. These methods enhance privacy protection and detection performance by integrating advanced cryptographic and machine learning techniques. Meanwhile, [33] also proposed the PEFL framework, which employs perturbation encoding and long short-term memory autoencoders to protect privacy and designs the FedGRU algorithm for intrusion detection. [34] proposed the P2TIF framework to protect confidential information in IIoT environments and identify cyber threats. It includes a blockchain module to ensure secure communication of data and a deep learning module that uses DVAE technology to resist inference attacks, followed by threat detection using A-DGRNN.

In recent years, IDS based on federated learning (FL) have attracted widespread attention [35]. FL, as a distributed machine learning approach, eliminates the need for sharing data samples and emphasizes data heterogeneity, availability, and privacy protection while reducing model bias, making it a highly promising solution for IIoT [36]. FL also provides a distributed framework that ensures privacy protection and promotes collaboration among multiple parties, enabling iterative training of machine learning models for IoT applications.

A blockchain-based DDoS attack detection framework using FL is proposed in [37]. It detects DDoS attacks in a distributed environment through data preprocessing (including data validation, feature selection, and normalization), local model training (using LSTM models and the Adam optimization algorithm), and model aggregation (employing centralized or distributed federated learning techniques). This framework aims to improve detection efficiency and accuracy while adapting to the distributed nature of blockchain networks.

However, these methods are facing several challenges in practical applications. First, in the process of detecting DDoS attacks on blockchain networks, it is not easy to find suitable

central server resources, mainly because resource-constrained edge devices cannot provide sufficient computing and storage capabilities to support large-scale model aggregation and optimization. Second, the failure of the central server may lead to the interruption of collaborative training, which in turn affects the stability and reliability of the entire federated learning system. In addition, as the number of participating devices increases, the scalability of the system and communication bottlenecks are becoming more and more prominent. These problems not only increase communication overhead but also may lead to a significant increase in training delays, thus posing a severe challenge to real-time performance and efficiency.

E. Blockchain Technology-Based Detection Methods

DDoS detection methods integrated with blockchain technology are gradually becoming a research direction. The decentralized and tamper-proof characteristics of blockchain provide new ideas for attack detection. For instance, blockchain-based smart contracts can establish a trust mechanism among distributed detection nodes, thereby enhancing the robustness of detection. In addition, blockchain technology can also be utilized to store attack features and historical data. For example [38] further discusses the application of blockchain in IoT, safeguarding device data through hash values and timestamps, and enabling the recovery to a secure state after network intrusions. [39] proposed a blockchain-based Link-Flooding Attack (LFA) detection scheme, which deploys blockchain technology on routers and servers within and around the protected area to record and share traceroute information, thereby achieving efficient detection of LFA.

Current researchers have proposed various approaches that integrate blockchain and machine learning techniques for detecting and defending against DDoS attacks. On one hand, traditional machine learning methods such as K-Nearest Neighbors (KNN), decision trees, and random forests have been demonstrated to effectively detect DDoS attacks [40]. Blockchain technology is utilized for securely storing blacklisted IP addresses, and smart contracts are employed to automatically block and unblock attack traffic. On the other hand, deep learning models like LSTM are applied to analyze network traffic from IoT edge devices, combining blockchain technology to achieve mechanisms for attack detection and defense [41]. Researchers in [42] have proposed a blockchain-based distributed machine learning framework that leverages collaboration among edge nodes and incentive mechanisms through smart contracts to enhance the efficiency of DDoS attack detection and model optimization capabilities. [43] introduces a DDoS detection framework integrating blockchain, smart contracts, and fuzzy neural networks. It protects privacy by hosting collaborators on private blockchains and avoids data sharing among them. The fuzzy neural network within the smart contract detects abnormal data, and the results are uploaded to a public blockchain for access by all collaborators, demonstrating high detection accuracy and recall rates. [45] presents a blockchain-supported adaptive DDoS detection and classification method. By integrating multiple deep learning models (GRU, CNN, LSTM, DNN, and SVM) and

introducing a dynamic weight adjustment mechanism, it significantly enhances the performance of DDoS attack detection and classification. [44] proposes a blockchain-supported Software-Defined Networking (SDN) framework (BSDN-HMTD). It employs deep learning techniques to detect DDoS attacks and strengthens network security and robustness through user authentication, flow validation, and dynamic Moving Target Defense (MTD) techniques.

However, in distributed systems such as blockchain networks, although the integration of blockchain and machine learning shows great potential in detecting DDoS attacks, it also faces many challenges. On the one hand, the transaction processing speed and storage capacity of blockchain are limited. In the scenario of high-traffic DDoS attacks, it may not be able to process and store a large amount of attack data in time, thereby increasing detection latency. On the other hand, machine learning models need to be updated regularly to adapt to new attack patterns. In distributed systems, how to efficiently update and synchronize model parameters is an urgent problem to be solved. In addition, real-time detection of DDoS attacks requires the system to respond quickly, but the transaction confirmation time and data synchronization time of blockchain may have a negative impact on the system's real-time performance.

III. CHALLENGES FOR DDoS ATTACK DETECTION IN BUSINESS BLOCKCHAIN NETWORKS

A. Insufficient Data Monitoring Coverage

1) *The Decentralization Dilemma of DDoS Attack Detection in Blockchain Networks:* In blockchain networks, the distributed architecture, while offering the advantages of decentralization and tamper resistance, also presents significant challenges for the detection of DDoS attacks. Given the extensive and independent distribution of nodes within blockchain networks, malicious traffic can easily be concealed within legitimate network flows, making it extremely difficult to distinguish between malevolent and benign traffic. This obfuscation not only increases the complexity of detection but also leads to a higher rate of false positives, thereby compromising the accuracy and reliability of detection systems. Moreover, the decentralized nature of blockchain networks further restricts the application of traditional centralized detection methods. In a decentralized environment, the absence of a unified control center to coordinate and implement detection strategies renders traditional single-point-based detection technologies inapplicable to blockchain networks. For instance, attackers can exploit the anonymity of blockchain nodes to launch attacks simultaneously from multiple nodes, thereby concealing their true identities and attack trajectories. This anonymity significantly diminishes the effectiveness of traditional tracking and identification methods when confronting DDoS attacks in blockchain networks, making it difficult to accurately locate the attack sources and implement effective defensive measures. Therefore, designing efficient, accurate, and adaptable DDoS attack detection mechanisms within the distributed environment of blockchain networks has become an urgent issue in the field of blockchain security research.

2) *Non-IID Data Distribution and Data Heterogeneity:* In blockchain networks, data typically exhibit Non-IID (non-independent and identically distributed) characteristics, with significant differences in the data distributions across various nodes. This data distribution characteristic poses substantial challenges for traditional machine learning models based on centralized training, as they often assume that data are IID. As a result, the generalization capability of detection models is compromised. For example, in blockchain networks, different nodes may process different types of transactions or traffic, leading to increased diversity in attack features.

The data distribution characteristics in blockchain networks show significant Non-IID features, which pose severe challenges to traditional machine learning models. Specifically, there are considerable differences in the data distributions across different nodes. These differences are not only reflected in the volume of data but also in the type, structure, and feature distribution of the data. For instance, some nodes may primarily handle high-frequency transaction data, while others may focus on smart contract execution or cross-chain interactions. This diverse data distribution makes it difficult for machine learning models based on centralized training to adapt effectively, as these models typically assume that data are IID and thus fail to fully account for data heterogeneity and complexity during the training process.

Moreover, the Non-IID data distribution characteristic further weakens the generalization capability of detection models. In blockchain networks, different nodes may face different types of attacks. For example, some nodes may be more susceptible to DDoS attacks, while others may face the risk of smart contract vulnerability exploitation. The diversity of attack features increases the complexity of detection models, making it difficult for a single centralized model to capture all potential attack patterns. For instance, an attack feature present on one node may not exist on another node or may appear in a different form. This heterogeneity in data and attack features requires detection models to have stronger adaptability and generalization capability in order to maintain high accuracy and low false-positive rates across different data distributions and attack scenarios. Therefore, designing machine learning models that can adapt to Non-IID data distributions within the distributed environment of blockchain networks has become an important research topic in the field of blockchain security. This not only requires the development of new algorithms to handle data heterogeneity but also necessitates exploring how to achieve collaborative training and optimization of models within a distributed architecture, thereby enhancing the overall performance and reliability of detection models.

B. The Real-Time and Scalability Challenges of DDoS Detection in Large-Scale Blockchain Networks

With the continuous expansion of the business chain network scale, DDoS attack detection systems are facing increasingly stringent requirements for real-time performance and scalability. In large-scale network environments, the volume of traffic data grows exponentially, and detection models must efficiently

process and analyze vast amounts of data within an extremely short period of time to minimize detection latency and ensure that attacks are identified and responded to in a timely manner. Meanwhile, the dynamic nature of the supervision chain, such as the frequent addition and exit of nodes, further complicates the detection system and poses significant challenges to its real-time performance and adaptability. In the actual deployment and application of supervision chains, the performance bottlenecks of blockchain technology have become a key obstacle to their large-scale application.

Specifically, limitations in transaction processing speed and storage capacity severely restrict the operational efficiency and scalability of supervision chain. On the one hand, although the consensus mechanism of blockchain ensures the security and immutability of data, it often comes at the cost of transaction processing speed. For example, when dealing with high-throughput network traffic data, the low transaction speed of blockchain can lead to data backlogs, preventing real-time attack detection and recording, thereby affecting the timeliness of detection. On the other hand, as the volume of regulatory data continues to rise, the storage capacity of blockchain is also reaching its limits. Distributed ledgers need to store large amounts of historical data, which not only increases the storage burden on nodes but also may cause network congestion and synchronization delays, thereby affecting the overall system's operational efficiency. Therefore, how to optimize the performance of blockchain while retaining its security and decentralization characteristics to meet the needs of large-scale applications has become one of the core issues that urgently need to be solved in the current research field of supervision chain.

C. The Tradeoff Between Privacy and Supervision

The application of the supervision chain in network security needs to not only meet the regulatory authorities' requirements for data transparency and traceability but also protect the data privacy of network participants. However, finding a balance between these two aspects is a significant challenge faced by the supervision chain. On the one hand, regulatory authorities need to obtain sufficient information to identify and prevent cyberattacks, ensuring the safety and stability of the network environment. For example, when detecting DDoS attacks, the supervision chain needs to record detailed characteristics of the attack traffic for tracing and analysis. On the other hand, the data privacy of network participants must not be overlooked. Although blockchain technology provides a certain level of privacy protection through encryption mechanisms, excessive data sharing may lead to the leakage of users' sensitive information in some cases. Therefore, designing mechanisms that can satisfy regulatory needs while protecting privacy is an urgent issue that needs to be addressed in the practical application of the supervision chain. This requires not only technological innovation, such as the application of privacy-preserving technologies like zero-knowledge proofs and homomorphic encryption, but also in-depth exploration at the legal and policy levels to ensure the compliance and acceptability of the supervision chain.

IV. ATTACK TYPE

This section introduces the types of DDoS attacks in blockchain networks.

A. DDoS Attack in Blockchain Data Layer

DDoS attacks on the blockchain data layer are malicious activities that interfere with data storage and integrity by sending a large number of invalid or low-value transactions, injecting false data, and generating invalid blocks. These actions consume the storage and computing resources of nodes, leading to decreased network performance and difficulties in confirming normal transactions.

B. DDoS Attack in Blockchain Network Layer

The Blockchain Network Layer DDoS attack model focuses on the blockchain network layer, specifically using blockchain's P2P (peer-to-peer) architecture for attacks. Attackers exploit vulnerabilities in P2P file sharing systems to access targeted objects through large numbers of normal users, or to disconnect some nodes from each other in an anonymous P2P network, thereby compromising the integrity and availability of the blockchain networks. This kind of attacks can cause the blockchain networks to fork, delay or fail transactions, affecting the decentralized nature of the blockchain and the trust mechanism.

C. DDoS Attacks in Consensus Layer

DDoS attacks at the consensus layer of blockchain typically occur by manipulating the consensus mechanism. Attackers may control a large number of nodes (e.g., by launching a 51% attack) or generate a large number of invalid blocks and transactions, thereby preventing the network from reaching consensus. This can lead to delayed transaction confirmations or even transaction rollbacks. Such an attack not only disrupts the stability and consistency of the network but also may affect the decentralized nature of the entire blockchain network, enabling attackers to manipulate transaction records or prevent the confirmation of legitimate transactions.

D. DDoS Attacks in Application Layer

DDoS attacks at the application layer of blockchain typically occur by launching a large number of requests against blockchain application programming interfaces (APIs) or smart contracts. Attackers exploit the openness and accessibility of these interfaces to send a vast number of malicious requests, which consume the computational resources and bandwidth of the application layer, leading to slow service responses or even system crashes. This type of attack not only increases the execution latency of smart contracts but also may prevent users from accessing blockchain applications normally, thereby severely affecting user experience and application availability.

E. DDoS Attacks in Protocol Layer

DDoS attacks at the protocol layer of blockchain typically exploit vulnerabilities or weaknesses in the blockchain protocol. Attackers may send a large number of maliciously constructed protocol messages to interrupt or disrupt communication between nodes, thereby undermining the normal synchronization and data transmission of the network. For example, attackers can exploit vulnerabilities in the P2P protocol to send a large number of forged node discovery or block propagation requests. When nodes process these invalid requests, they consume a significant amount of resources, which may ultimately prevent them from functioning properly and even lead to network partitioning. This type of attack directly affects the underlying communication mechanism of the blockchain network, posing a threat to the stability and security of the entire network.

F. DDoS Attacks in Node Layer

DDoS attacks at the node layer of blockchain typically occur by focusing attacks on specific nodes. Attackers may overwhelm the target nodes with massive malicious traffic or generate a large number of spurious requests, thereby exhausting the resources of the target nodes (such as CPU, memory, and bandwidth). This prevents the nodes from processing legitimate transactions or communicating normally with other nodes. Such an attack not only degrades the performance of the nodes but also may cause them to crash or become isolated, thereby affecting the decentralized nature and overall stability of the entire blockchain network.

G. DDoS Attacks in Smart Contract

The DDoS attack model against smart contracts targets blockchain-IoT systems based on smart contracts, specifically exploiting the automated nature of smart contracts. Attackers could exploit vulnerabilities in smart contracts or clog up the network by sending a large number of illegal requests, making it impossible for smart contracts to properly execute or process legitimate transactions. Such attacks can result in the failure of smart contract execution, affect the business processes of blockchain-IoT systems, and may even lead to financial losses.

V. METHOD

In this part, we will introduce the following aspects: the design of the model architecture as well as the workflow, network traffic preprocessing technology and the training method of the model.

A. Model Architecture

Our method consists of several entities: supervision chain, clients, and business network. In our federated learning framework, model aggregation is performed by supervision chain, which includes a leader node and several follower nodes. Follower nodes establish communication connections with client. clients are not only used to upload and update parameters during model training, but also used to analyze network traffic generated by the business networks during model detection and record it

on the blockchain as transactions. The leader node is responsible for verifying transactions and proposing new blocks. And help maintain the consistency of model parameters across nodes of the Directed Acyclic Graph (DAG) supervision chain, coordinating parameter aggregation and model updates throughout the federated learning process. Leveraging the high throughput and efficiency inherent to DAG-structured blockchains, the supervision chain is capable of rapidly recording model parameters and vast numbers of detected DDoS attack events. This enables a more agile and responsive detection mechanism, enhancing the overall security and robustness of the system.

1) *Workflow of DDoS Attack Detection Model:* The workflow of this model is shown in Fig. 2:

i) Each client in business chain have a labeled data set and a set of business blockchain nodes that it monitors.

ii) Each client in business chain uses local datasets for training.

iii) Each client in business chain completes the current round of training and obtains the local model data.

iv) The client in business chain that completes this training round will upload the local model data to the expected follower node on the supervision chain, and the follower node will record the model parameters as transactions on the supervision chain through "Data Upload" smart contract.

v) The leader node obtains the aggregation model through our proposed aggregation algorithm by reading and analyzing the model parameters of the clients node received on the supervision chain. The leader node records the aggregated model parameters on the DAG ledger through a "Data Upload" smart contract and shares them with the follower nodes. You can see the details in Algorithm 1.

vi) The followers node reads the aggregated model parameters through the "Data Query" smart contract from DAG ledger, sends them to the corresponding clients, and updates the model parameters of the clients.

vii) After multiple rounds of training, the classification accuracy of the model reached a reasonable standard, and the model training was completed. Each client node will collect the network traffic of its assigned business blockchain nodes through the traffic collection device, and analyze these traffic to detect potential DDoS attacks.

viii) The client saves the latest network traffic check result. This information is not only used to update local training datasets, It is also uploaded via the client node to its corresponding follower node on the DAG blockchain and stored on the DAG chain as transactions through smart contracts called "data uploads," sharing information on discovered attack behavior with other nodes.

Follower nodes in the supervision chain are responsible for updating model parameters during the federated learning process. They record the network traffic generated by the business blockchain nodes as transactions on the supervision chain ledger. These follower nodes interact with the supervision chain leader and client nodes in the business chain. They obtain locally trained model parameters from the client nodes in the business chain and upload them to the leader node, which aggregates these parameters using the proposed asynchronous federated learning algorithm. The follower nodes ensure that only model

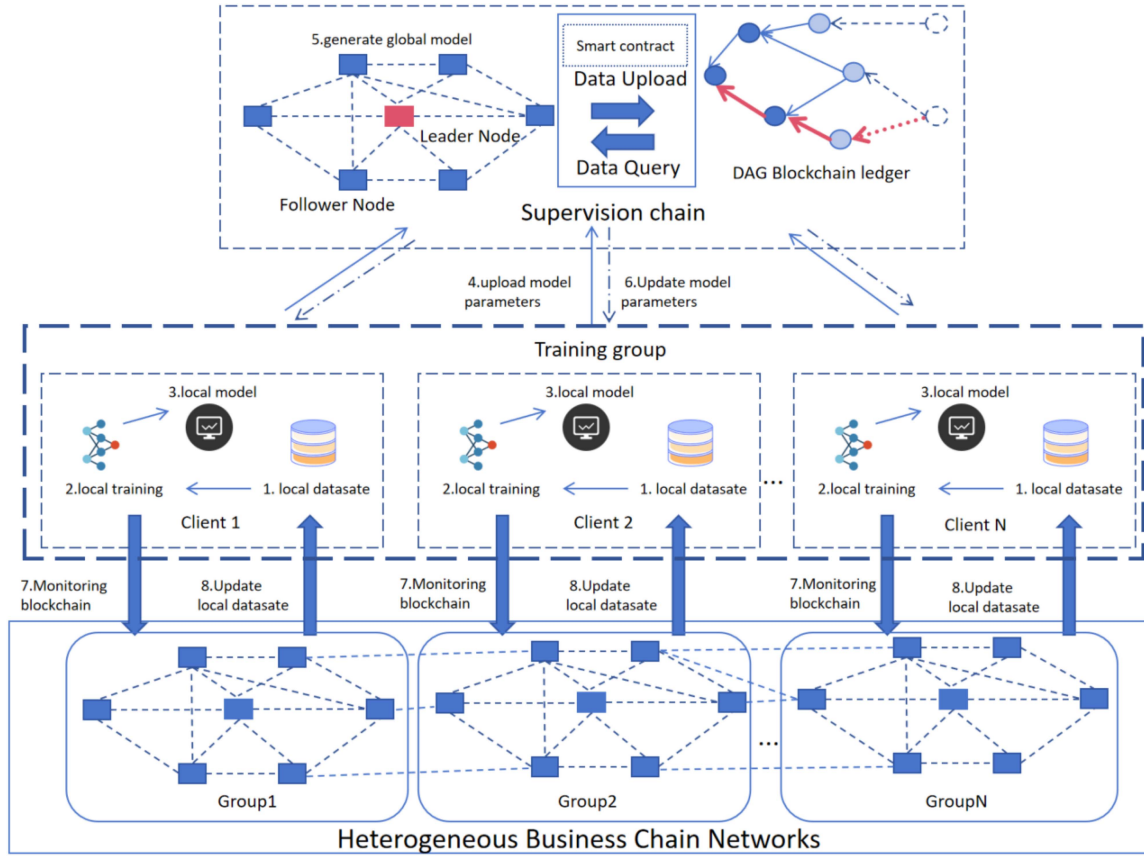


Fig. 2. Working process of BAFL DDos attack detection system.

parameters are shared, not the raw network traffic data from the business chains, thereby protecting the data privacy of the business chains.

Client nodes corresponding to the business chain monitor and collect network traffic data from the business blockchain nodes. They preprocess this data and use it to train local models. These client nodes send their locally trained model parameters to the supervision chain for aggregation and further processing. Meanwhile, they also receive updated global model parameters from the supervision chain to continue training in subsequent rounds. After training is completed, the client nodes corresponding to the business chain act as regulators to conduct security monitoring of the business chains they oversee.

Our aggregation node of federated learning consist of a DAG structured blockchain, and the implementation of consensus relies on the process of selecting and confirming transactions among the nodes in the DAG network, where the tips transaction block refers to the most recent collection of transactions that have not yet been confirmed by subsequent transactions. Nodes reach a consensus by selecting the Tips trading block with the highest credibility that satisfies a particular selection algorithm, such as the highest transaction fees or the greatest network influence, and these choices reflect a common recognition of transaction order and validity in the network. As more transactions join, they reference these Tips trading blocks, gradually building an interdependent network structure to achieve a common

understanding of the history and current status of the transaction. This graph-based structure allows transactions to be confirmed in a non-linear manner, improving the throughput of the system and enabling a decentralized consensus mechanism through the joint selection and confirmation of Tips trading blocks by nodes in the network.

The DAG structure supports the parallel confirmation of multiple transactions, effectively avoiding the sequential processing bottleneck in traditional blockchains. Its throughput increases linearly with the number of nodes, making it suitable for large-scale networks. Meanwhile, DAG's efficient transaction recording capability, such as processing thousands of transactions per second, reduces the latency of uploading model parameters and attack events to the blockchain, thereby supporting real-time detection and response.

In the process of model aggregation, the aggregator determines the weight of the parameter in the model aggregation according to the old and new degree of the parameter transmitted by the client and the accuracy rate of the client model. In the process of the first aggregation, the aggregator first aggregates the collected client parameters to obtain the pre-aggregation model of the round and its accuracy, and then compares the model accuracy of each client with it. When the accuracy of the pre-aggregation model is high, the parameters of the pre-aggregation model dominate; otherwise, the parameters of the client model dominate. As the result, the older the client

model parameters, the smaller the proportion in the aggregation process, we implemented this by introducing a time factor. This is described in detail in the part C of Section V.

The time factor indicates that in the maximum experimental time range, the longer the time delay, the smaller the time factor, resulting in a smaller model proportion. If the delay exceeds the maximum delay range, the client model can be ignored.

During the training of the DDoS attack detection model, clients in federated learning only upload model parameters to the server, without transmitting raw training data. This mechanism ensures the privacy of the network traffic data in business chains, as different business chains are monitored by separate clients and data between chains are isolated from each other. Moreover, we have automated the processes of data verification and sharing through the use of smart contracts, thereby further enhancing the security and trustworthiness of the data. In the actual deployment scenarios of supervision chains, smart contracts are designed to permit only specific nodes to access and process data. Under this architecture, only regulatory authorities and certain business chain nodes are able to obtain certain sensitive data, thereby realizing a precise and meticulous access control mechanism. For example, in the IoT environment, the data generated by devices may contain users' private information. Through blockchain technology and smart contracts, the privacy of this data can be ensured during transmission and processing, preventing data leakage and misuse.

In the actual deployment scenarios of supervision chains, smart contracts are designed to permit only specific nodes to access and process data. Under this architecture, only regulatory authorities and certain business chain nodes are able to obtain certain sensitive data, thereby realizing a precise and meticulous access control mechanism.

This approach of combining the DAG structure with federated learning not only enhances the performance and efficiency of BAFL but also strengthens data privacy protection and model robustness, making it a powerful solution for protecting blockchain networks from DDoS attacks. It is particularly effective in processing large-scale, dynamically changing network data, providing high-precision, low-latency detection results. Firstly, from the perspective of data privacy protection, federated learning effectively safeguards data privacy by sharing only model parameters instead of raw data. The supervision chain of the DAG structure further ensures the immutability and traceability of these parameters, thereby enhancing the credibility of data privacy protection. Secondly, in terms of efficient data sharing and synchronization, the efficiency of the DAG structure enables rapid uploading and synchronization of model parameters. This is especially crucial for handling high-traffic network data, as the DAG structure can process thousands of transactions per second, significantly reducing the latency of data uploading and synchronization. Moreover, regarding model robustness and scalability, the asynchronous nature of federated learning makes the model training process more flexible, better able to handle client failures and network latency. The supervision chain of the DAG structure ensures real-time updating and sharing of model parameters, thereby enhancing model robustness and scalability. Finally, in the aspect of real-time detection and response, this

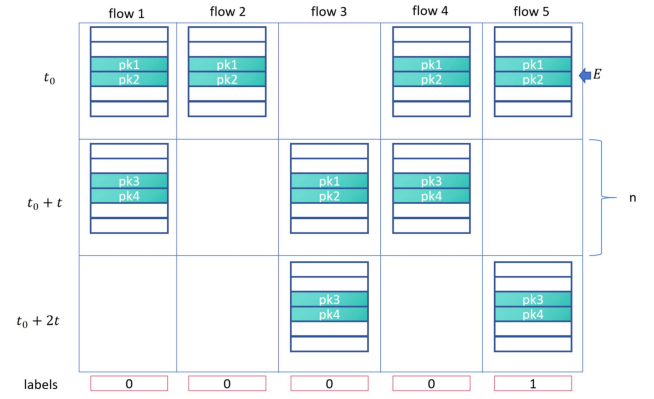


Fig. 3. The process of data preprocessing.

combination enables BAFL to quickly update and synchronize model parameters, ensuring the timeliness of detection results. Thus, when dealing with dynamically changing network traffic data, it can more promptly respond to potential DDoS attacks.

B. Real-Time Network Traffic Preprocessing

Network traffic comprises data transfers between endpoints. Because the communication channel is shared, packets from different data flows are multiplexed, or split for transmission, for the purpose of transmission. This suggests that handling traffic for a live presentation to an NIDS is quite different from handling a static dataset that contains all of the flows [46].

We use tool designed in paper [46] to develop our online detection, which divides the traffic flows into sub-flows based on time frames and transforms them into array-like data structures taken from network traffic traces of a dataset.

Network traffic consists of data flows between ends, as shown in Fig. 3. Due to the shared nature of communication links, packets from different data streams are multiplexed, resulting in packets from the same data stream being separated for transmission. This means that real-time traffic is being processed.

In the feature extraction stage, given a traffic tracking file from the data set and a predefined time window with a length of t seconds, collect all packets ($pk1, pk2, \dots$) from the file with the capture time between t_0 , the capture time of the first packet, and time $[t_0 + t]$. The algorithm extracts f attributes from each packet.

In the data preprocessing stage, the attributes of packets belonging to the same bidirectional communication stream are grouped in chronological order to form a matrix of shape width and length $[n, f]$, where f is the number of features and n is the maximum number of packets collected for each stream in the time window during the parsing process. Streams greater than n are truncated, while shorter streams are filled with zeros at the end of the next stage after normalization. In the time window $[t_0 + t, t_0 + 2t]$, etc. The same operation is repeated for the packet until the end of the file. The output of this process can be viewed as a two-dimensional array of samples E . A row in the array represents a sample of the packet captured in the same

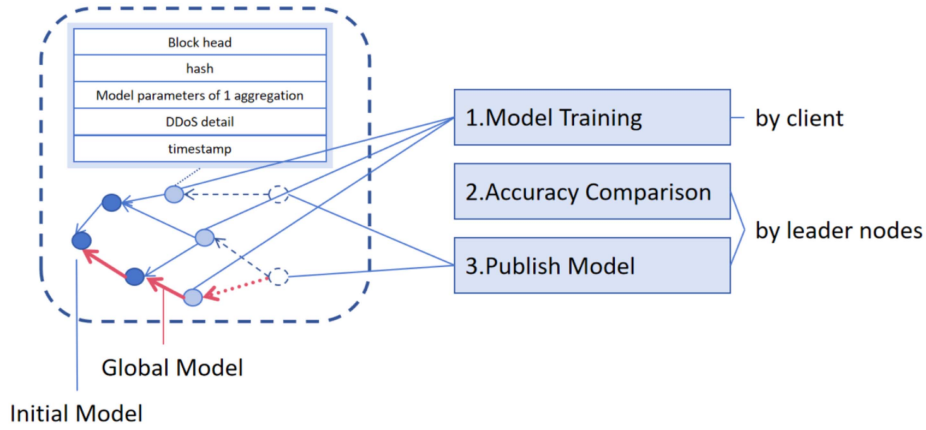


Fig. 4. BAFL training procedure.

time window, while a column represents a sample of the packet belonging to the same bidirectional stream.

In the normalization and filling stage, each attribute value is normalized to a scale $[0,1]$, and the sample is filled with zero, so the fixed length of each sample is n , because a sample with a fixed length is the requirement that CNN can learn on a complete sample set. Each non-empty element of array E is a compact graphical representation of a sample. Within each E element, the colored rows are the packet in the form of f standardized attributes, while the white rows represent zero padding.

Each E is labeled by matching its stream identifier id with the label provided by the original dataset. This also means that the value of the label is constant along each column of the array E .

In the input layer, each traffic stream is reshaped in seconds into a two-dimensional matrix of packet characteristics, enabling CNNs to learn correlations between packets of the same stream. This first layer takes a flow flow represented by a matrix of size $n \times f$ as the input to the neural network.

Each input matrix is operated by a convolution layer with k convolution nuclei, each of size $h \times f$, where h is the length of each convolution kernel. Our neural network model is trained to classify network traffic.

C. Model Training

We randomly and evenly distributed the preprocessed data to each client p , allowing local data on different clients to cover different feature spaces or categories. Clients with higher accuracy may have learned more generalizable features or their local data distributions may be closer to the global task requirements. By assigning higher weights to models with higher accuracy and introducing a temporal factor to reduce the aggregation weights of outdated models, we can accelerate the convergence of the global model and potentially enhance its performance across the overall data distribution. The asynchronous federated learning algorithm based on accuracy is shown in Algorithm 1.

As for each client p , we divide the collected attack network traffic data into training set and test set. After receiving the data, the client uses the CNN model for training. The trained client sends the model parameters w , model accuracy acc , and losses

to its corresponding follower node, which packages this information into the transaction and uploads it to the DAG blockchain for data sharing.

The leader node reads the model parameters of follower nodes that have completed the current theoretical training from the DAG chain. According to the accuracy of the client model corresponding to each follower node, the leader node weights the parameters so that the model parameters of the client with higher training accuracy account for a larger proportion in the aggregation. After the leader node aggregates these parameters, it packages the aggregated parameters w into a transaction and publishes it on the DAG blockchain, sharing the model parameter information with the follower nodes, who send these parameters to the client for the next round of training. The BAFL training procedure is shown in Fig. 4.

1) *Local Model Training*: The training sample is evenly distributed to each client, and the number of client is P . After receiving the training samples sent by the training data memory, we randomly divided the training samples into P parts and gave them to each client. After each follower node receives the training sample $\mathcal{D}_{tp} = (d_{k, \frac{n}{p}-1}, \dots, d_{k, \frac{n}{p}})$, using the t round of training assigned to its own data.

We collect real-time network traffic data from a blockchain network, pre-process it, and apply it as input to our convolutional neural network (CNN) model to facilitate online attack detection. Traffic data is divided into two categories: malicious (DDoS attack) traffic and benign traffic. We aim to optimize the CNN model to reduce complexity and computation time so that the model can be effectively deployed on resource-constrained devices. In order to achieve this goal, we design a lightweight supervised learning system, the core of which is a well-designed CNN for efficient attack detection.

The structure of our neural network is shown in Fig. 5:

Input layer: We build a spatial representation in which every traffic stream is transformed into a two-dimensional packet characteristic matrix in a matter of seconds. To make it possible for the CNN to discover correlations between the same stream's packets. As a result, a flow stream represented by a matrix f of size $n \times f$ is fed into this first layer. F is made up of n distinct package vectors, therefore $F = pkt1, \dots, pkt n$. where

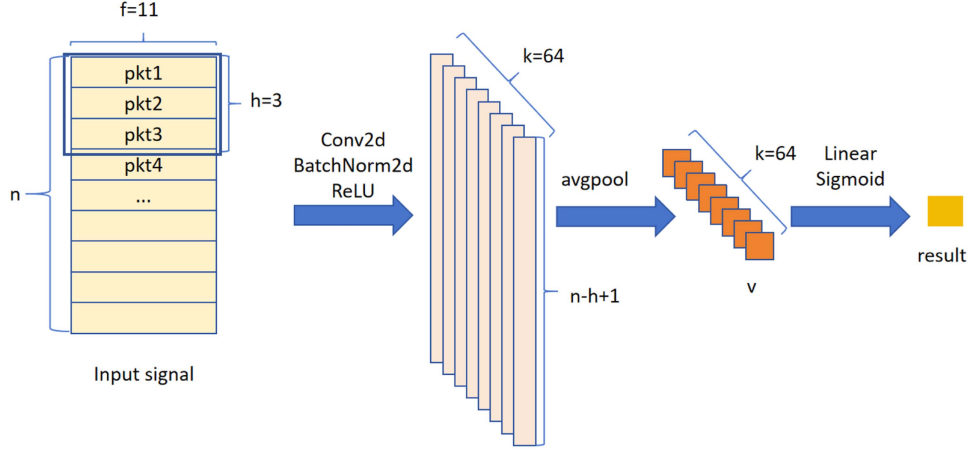


Fig. 5. The structure of our neural network.

each packet vector has a length of $f = 11$ features and pkt_n is the n TH packet in a stream.

CNN layer: As shown in Fig. 5, each input matrix F is operated by a convolution layer, in which the size of k filters is $h \times f$, and h is the length of each filter. In our model, we set $k = 64$ to make the data features pass through multiple channels, so that the characteristics of DDoS attacks can be extracted more comprehensively. Each filter, also known as the kernel or sliding window, performs convolution with step 1 to extract and learn local features that contain useful information for detecting DDoS and benign flows. Each filter generates an activation map of magnitude $(n - h + 1) \times k$, such that $a_k = \text{ReLU}(\text{Conv}(F)W_k, b_k)$, where W_k and b_k are the weight and bias parameters of the KTH filter learned during the training phase. We add BatchNorm2d after the convolutional layer of the convolutional neural network for data normalization processing, so that the data will not cause unstable network performance due to large data before Relu. We then use the corrected linear activation function $\text{ReLU}(x) = \max(0, x)$. All activation graphs are stacked to create an activation matrix A of size $(n - h + 1) \times k$, which gives us $A = [a_1 | \dots | a_k]$.

Average pooling layer: We use an average pooling method where we sample down the first dimension of A , which represents the time nature of the input. A pool of size m produces an output matrix of size $((n - h + 1)/m) \times k$ containing the maximum m activation for each learning filter, such that $c_0 = [\text{avg}_1 | \dots | \text{avg}_k]$. In this way, the dimensionality of the training data is reduced, the redundant information is removed, the features are compressed, the network complexity is simplified, the computation is reduced, the memory consumption is reduced, and too much high-dimensional information is avoided. c_0 then flattens to generate a one-dimensional feature vector v that will be input to the detection layer.

Detection layer: v is input to a linear fully connected layer. The main function of the Linear fully connected layer is to map the feature space computed from the previous layer to the sample label space. That is, the feature representation is integrated into one value, which has the advantage of reducing the influence

of feature positions on classification results and improving the robustness of the entire network. This output x is passed to the S-type activation function such that $\sigma(x) = 1/(1 + e^{-x})$. This limits activation to a value between 0 and 1, thus returning the probability $p \in [0, 1]$ that a given stream becomes a malicious DDoS attack. If $p > 0.5$, the traffic is DDoS. Otherwise, the traffic is benign.

When the local model is trained for the second time or more, we optimize the parameters of the local model based on the local model trained in the current round. Construction loss function

$$\mathcal{L}^{pt} = \sum_{i=1}^n l(y_i, \hat{y}_{p,i}^{t-1}) + \Omega(f_{pt}) \quad (1)$$

It is minimized to represent the predicted value when the locally trained model p trains the sample in the round of t training. After the training is completed, the parameters of CNN network parameter will be sent to the server. Regularized term $\Omega(f_{pt})$ controls model complexity and prevents overfitting and weight decay.

2) Aggregation Process: After the local training model completes the r th round of training, clients send its r th round of training parameters w_r^p and r th round of accuracy acc_r^p to the leader node to train the global model. The leader node uses fedprox aggregation algorithm to get the preliminary aggregation parameters of round r w_r^p . We set the difference between the time when the server receives the data from client p and the time when client p sends the r round of model parameters to the server as t_r^p , the longest waiting time of the server is max_t , and the calculation formula of the time-weighted parameter of client p ct_r^p is as follows

$$ct_r^p = \begin{cases} (1 - (t_r^p - max_t) / max_t) / 100 + C, & 0 < t_r^p < max_t \\ 0, & otherwise \end{cases} \quad (2)$$

Here C is the adjustment parameter. The leader node then recalculates the aggregate parameters by determining how accurate each client's model parameter is compared to client's accuracy. When the accuracy of the client's model is higher, the model parameters of client will play a dominant role in the adjustment process of the aggregation parameters, and the

Algorithm 1: Asynchronous Federated Learning Algorithm
Based on Accuracy.

```

for each round  $r = 1, 2, \dots$  do
  Leader node executes: initialize  $w_r^p$ 
  get  $ct_r^p \leftarrow t_r^p$ 
  for each client  $p$ , in parallel do
     $w_{r+1}^p \leftarrow ClientUpdate(p, w_r^p)$ , calculate  $acc_{r+1}^p$ 
    if  $acc_r^p > acc_{r+1}^p$  then
       $w_{r+1}^p = w_r^p \times (1 - ct_r^p) + w_{r+1}^p \times ct_r^p$ 
    else
       $w_{r+1}^p = w_{r+1}^p \times (1 - ct_r^p) + w_r^p \times ct_r^p$ 
    end if
  end for
  Calculate the aggregation parameters:
   $w_{r+1} \leftarrow FedProx$  all  $w_{r+1}^p$ 
  Follower node update( $p, w_r^p$ ): // Executed on client  $k$ 
  for each local epoch do
     $batches \leftarrow$  data split into batches
    for Batch  $b$  In Batches do
       $w_r^p \leftarrow w_r^p - \eta \nabla L^{pt}(w_r^p, b)$ 
    end for
  end for
end for
return  $w_r^p, acc_r^p$ ;

```

TABLE II
HYPERPARAMETER SETTING

Parameter	Value	Description
Number of Clients	20	Total number of clients participating in federated learning
Local Training Epochs	5	Number of local training iterations per communication round for each client
Batch Size	16	Batch size used for local training
Learning Rate	0.001	Initial learning rate for the Adam optimizer
Regularization Coefficient	0.0001	L2 regularization term to prevent overfitting

effectively reduces the impact of outdated model parameters on the current training process, thereby enabling asynchronous federated learning. This strategy not only accelerates the speed of model aggregation but also enhances the robustness of the model training process. Even if some clients fail, model training can still continue and can quickly integrate the latest parameter updates from different clients, ensuring the continuity and efficiency of the training process and maintaining high quality and efficiency. In addition, to fully leverage the high-performing follower nodes, our leader node performs a secondary adjustment of the model parameters based on the accuracy of each client after the initial aggregation in each round, further speeding up the aggregation process.

model parameters after the update of the r round client p are $w_{r+1}^p = w_{r+1}^p \times (1 - ct_r^p) + w_r^p \times ct_r^p$. On the contrary, when the aggregation model accuracy of the leader node is higher, the aggregation model parameters of the leader node will dominate the aggregation parameter adjustment process, and the model parameters after the r round client p update are $w_{r+1}^p = w_r^p \times (1 - ct_r^p) + w_{r+1}^p \times ct_r^p$. After calculating the new weighted parameters of all client, perform the next aggregation to obtain the final aggregation parameters and send them to supervision chain.

Clients may encounter various faults or anomalies during the model training process, such as operating system failures, memory overflows, and insufficient storage space. These technical issues may cause clients to fail temporarily or permanently, thereby affecting the continuity and stability of the federated learning process. In addition, the uneven quality of network connections between clients makes the inconsistency of data transfer rates a key issue. Factors such as network latency, bandwidth limitations, and unstable connections may cause some clients to be less efficient than others in sending or receiving model updates. This instability of clients can lead to failures at any time, preventing the aggregation model from obtaining the latest parameter updates in a timely manner, which in turn affects the real-time and accuracy of model training. Secondly, some clients may achieve higher accuracy than the leading node during training, while others do not. In this case, traditional aggregation methods may mask the advantages of these high-performing models.

To address these challenges, this study introduces a time decay parameter and proposes an innovative method. This method

VI. EXPERIMENT AND EVALUATION

In this study, our core objective was to conduct a comprehensive performance evaluation of our proposed BAFL approach. Specifically, this evaluation aims to explore the performance of the BAFL approach on several key dimensions. First, the accuracy of its global model in identifying DDoS attacks. Secondly, its robustness is tested in the face of different proportions of node failures. Thirdly, the convergence rate in the federated learning process is evaluated. Finally, the scalability of the system when processing large-scale data is evaluated. Through these comprehensive tests, we aim to gain a deeper understanding of the effectiveness and potential of the BAFL approach in practical applications.

Our model was trained on a personal computer configured with 13th Gen Intel(R) Core(TM) i9-13900HX, and CUDA 11.6 version. The parameter Settings in this study are detailed in Table II. All experimental results are based on the statistical average of 10 independent repeated experiments.

This model is validated with normal blockchain network traffic data and recent datasets CIC-DDoS2019 provided by the Canadian Institute for Cybersecurity of the UNB, Canada [47]. [47] consist of several days of network activity, normal and malicious, including DDoS attacks. The dataset is publicly available in the form of traffic traces in pcap format including full packet payloads, plus supplementary text files containing the labels and statistical details for each traffic flow. The attack types in CIC-DDoS2019 DDoS shown in Table III.

These datasets encompass both normal network traffic and a variety of DDoS attack traffic, such as IRC-based botnet

TABLE III
CIC-DDoS2019 DDoS ATTACK TYPES

Attack	Description	Transport
TFTP	An attack is constructed by mirroring the files sent to a Trivial File Transfer Protocol (TFTP) server in the direction of the fake IP address of the victim.	UDP
Syn Flood	Attack that uses a deluge of SYN packets to overwhelm the victim's resources by taking advantage of the TCP three-handshake mechanism	TCP
UDP Flood	The attack is designed to use as many tiny fake UDP packets as possible in order to overwhelm the victim's network capacity.	UDP
UDPLag	UDP traffic produced with the intention of hindering the victim's online gaming server connection.	UDP
WebDDoS	A brief DDoS assault on a web server on port 80, consisting of approximately 3100 packets.	TCP
SSDP	attacks based on the Simple Service Discovery Protocol (SSDP), which enables information transmission and reception over the UDP protocol between UPnP devices. When the victim's IP address is faked, the susceptible device responds with a UPnP message.	UDP
SNMP	Reflection amplification attacks employing the udp-based Simple Network Management Protocol (SNMP) to configure network devices.	UDP
DNS	DDoS assaults provide a response to an attack request to a server using the fake IP address of the victim in order to flood victims by employing certain UDP-based network services.	UDP

attacks and HTTP DDoS attacks utilizing LOIC and HOIC tools. They also feature a diverse range of traffic characteristics spanning the network, transport, and application layers, including packet length, protocol type, TCP flags, and window size. These datasets exhibit non-independent and identically distributed (non-i.i.d.) characteristics: attack traffic significantly differs from normal traffic in terms of features (e.g., higher packet rates and more concentrated IP address distributions in attack traffic), and different types of DDoS attacks also possess distinct traffic features (e.g., differences between network-layer and application-layer attacks). Moreover, the datasets are divided into time windows, and the traffic features within each window may vary due to the influence of network conditions, user behavior, and attack activities.

When these non-i.i.d. datasets are distributed across different clients, heterogeneity arises primarily due to the diversity of data distributions, the variety of data features, and the imbalance of data volumes. In the context of DDoS attack detection in service chain networks, the significant differences in features (such as packet rates and IP address distributions) between different types of attack traffic and normal traffic contribute to the diversity of data distributions. Additionally, variations in network environments across organizations, data collection time windows, and the volume of data provided by each client further exacerbate the diversity of data features and the imbalance of data volumes. The datasets used in the paper, which include multiple attack types

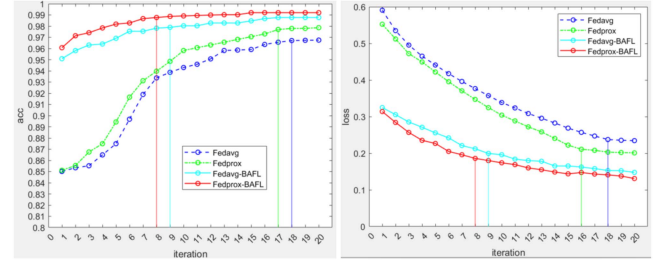


Fig. 6. Accuracy and loss of our model BAFL compared to traditional methods.

and diverse traffic features, are processed by time-window-based division and merging, all of which contribute to the evident heterogeneity of the data.

A. The Performance of the Global Model to Identify DDoS Attacks

Classification performance of BAFL models is measured in Precision, recall rate and F1-score. In machine learning and statistics, Precision, Recall, and F1-score are three important metrics for evaluating the performance of classification models, especially in binary classification problems. Here's what each means:

Loss function (loss): The loss function measures the difference between the model's predicted value and the actual value. It is an objective function in the process of model optimization and is used to guide the adjustment of model parameters to minimize prediction errors. Different machine learning tasks use different loss functions, and our classification task uses cross entropy loss.

Accuracy (acc): Accuracy refers to the proportion of all predictions in the model that are correct.

Precision: The accuracy rate refers to the proportion of the positive class that the model predicts is actually positive.

Recall rate: The recall rate refers to the proportion of positive class samples correctly identified by the model to all actual positive class samples.

F1-score: The F1-score is a harmonic average of accuracy and recall, and it tries to find a balance between the two metrics. When the accuracy and recall rate are equal, the F1-score reaches the maximum.

These metrics provide a comprehensive assessment of the model's performance, especially if the sample of positive and negative categories is unbalanced.

In this part, we used Fedprox-BAFL, Fedavg-BAFL, Fedavg, Fedprox and other methods to train the global model until the model converged. Our model underwent 20 iteration of aggregation during training. To evaluate the performance of these methods, we compared their Acc and Loss values over different iterations. The experiment was performed 10 times per experiment through the collaborative participation of 20 clients to ensure statistical significance of the results, and average indicators are reported in this section.

As shown in Fig. 6 and Table IV, under the same number of iterations, Fedprox-BAFL and Fedavg-BAFL models, through

TABLE IV
MODEL PERFORMANCE COMPARISON

	Precision	Recall	F1-score	Convergence time	LOSS	ACC
FEDAVG	0.97484	0.94544	0.97474	249.0s	0.2361	0.9654
FRDPROX	0.98019	0.98019	0.96078	184.58s	0.1800	0.9803
BAFL-FEDAVG	0.98048	0.99504	0.98770	92.0s	0.1478	0.9877
BAFL-FRDPROX	0.99014	0.99505	0.99259	69.3s	0.1310	0.9926

our quadratic aggregation algorithm, give greater weight to the model parameters of clients with higher accuracy during model aggregation, which makes our model more accurate than FEDAVG and FEDPROX, showing higher accuracy and lower losses. In particular, the acc of Fedprox-BAFL and Fedavg-BAFL models has reached convergence at the 8th and 9th iterations, and the convergence time is 69.3 s and 92.05 s, respectively. Acc converges at the 18th and 19th iterations of Fedprox and Fedavg, which are 184.583 s and 249.02 s, respectively. Similarly, the loss values of Fedprox-BAFL and Fedavg-BAFL models converged at the 8th and 9th polymerization, while Fedprox and Fedavg converged at the 16th and 18th polymerization, respectively. Compared with FEDAVG and FEDPROX, the improved model FedPROX-BAFL and FeDAVG-BAFL also achieved higher scores in Precision, Recall and F1-score.

Further, considering that the data set contains multiple attack types, Fedprox-BAFL method shows better performance than Fedavg-BAFL in heterogeneous data detection. Specifically, under the same number of iterations, Fedprox-BAFL method not only has higher accuracy and lower loss, but also has faster convergence of accuracy and loss value. These results show that the Fedprox-BAFL method can effectively improve the performance and convergence rate of the model when dealing with data sets containing multiple attack types.

The above experiments show that BAFL employs a dynamic weight allocation mechanism, assigning different weights based on the detection accuracy of client models. Models with better performance have a larger proportion in the aggregation process, thereby enhancing the accuracy of the global model. The Fedprox-BAFL method can more effectively utilize the parameters of high-precision models when dealing with complex datasets containing various types of attacks, thereby enhancing the overall performance of the model. Additionally, BAFL introduces a temporal decay factor, which reduces the impact of old model parameters on the aggregation results. This enables the model to quickly adapt to new data distributions and accelerates the convergence process. When processing dynamically changing network traffic data, the temporal decay factor can ensure that the model does not overly rely on old and potentially outdated parameters, thereby maintaining the real-time and accurate nature of the model.

B. Robustness

In this study, we aim to evaluate the performance of the proposed Fedprox-BAFL model in the face of node failure. Specifically, we randomly suspended the training of some clients to simulate mid-process client failures. We introduced random node failures in the 2nd to 5th rounds of model training, and

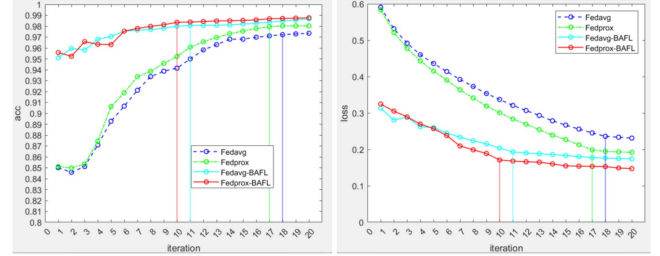


Fig. 7. Accuracy and loss of their model when 20% of clients are down.

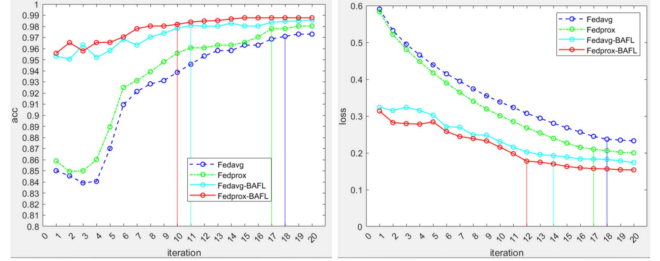


Fig. 8. Accuracy and loss of their model when 40% of clients are down.

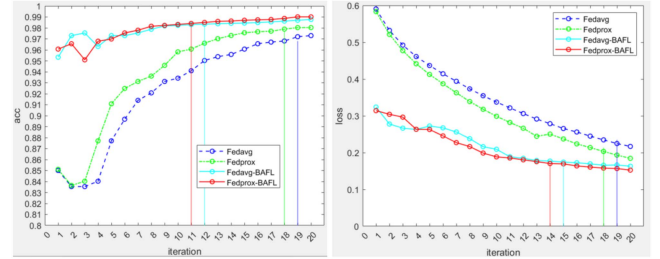


Fig. 9. Accuracy and loss of their model when 60% of clients are down.

compared and analyzed the performance of Fedprox-BAFL, Fedavg-BAFL, Fedavg, Fedprox models under different proportions of fault nodes (20%, 40%, 60% and 80%). As shown in Figs. 7–10, node failures lead to fluctuations in the accuracy and loss values of all models, and such fluctuations are aggravated as the proportion of faulty nodes increases. In addition, the increase in the number of faulty nodes also prolongs the number of iterations and the time required for model convergence.

In the presence of node failures, the aggregation process is affected by the uneven distribution of data. In federated learning, each node may have different data distributions. If the failure of some nodes leads to the loss of key data, the distribution of remaining data may be changed, thus affecting

TABLE V
MODEL PERFORMANCE COMPARISON WHEN 20% OF CLIENTS ARE DOWN

	Precision	Recall	F1-score	Convergence time	LOSS	ACC
FEDAVG	0.97040	0.98019	0.96078	219.6s	0.1781	0.9803
FRDPROX	0.98019	0.98019	0.97527	210.4s	0.2187	0.9745
BAFL-FEDAVG	0.98031	0.98514	0.98272	95.5s	0.1804	0.9877
BAFL-FRDPROX	0.98058	0.99999	0.99020	81.6s	0.1608	0.9902

TABLE VI
MODEL PERFORMANCE COMPARISON WHEN 40% OF CLIENTS ARE DOWN

	Precision	Recall	F1-score	Convergence time	LOSS	ACC
FEDAVG	0.97049	0.98016	0.97530	230.8s	0.2834	0.9738
FRDPROX	0.97058	0.98020	0.97537	221.1s	0.2729	0.9741
BAFL-FEDAVG	0.98031	0.98523	0.98301	98.3s	0.1739	0.9877
BAFL-FRDPROX	0.98061	0.99999	0.99019	91.2s	0.1663	0.9902

TABLE VII
MODEL PERFORMANCE COMPARISON WHEN 60% OF CLIENTS ARE DOWN

	Precision	Recall	F1-score	Convergence time	LOSS	ACC
FEDAVG	0.98465	0.95539	0.96980	236.6s	0.2802	0.9698
FRDPROX	0.98469	0.95545	0.96985	227.5s	0.2170	0.9705
BAFL-FEDAVG	0.98031	0.98514	0.98272	101.3s	0.1804	0.9877
BAFL-FRDPROX	0.98058	0.99999	0.99020	98.6s	0.1608	0.9902

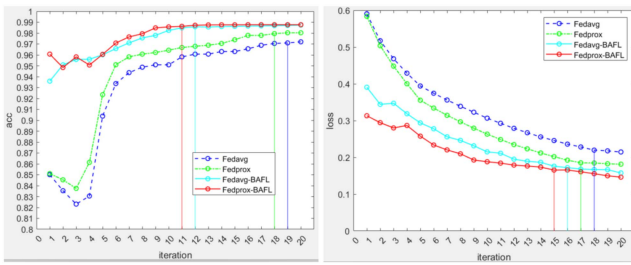


Fig. 10. Accuracy and loss of their model when 80% of clients are down.

the performance of the model in each iteration. The instability of model updating is also a key factor. Model updating depends on the data and computing power of each node, and node failure may cause the updating process to become unstable, especially when the number of nodes is small, and the update of a single node has a more significant impact on the global model. In addition, node failure may change the communication mode of the network, affect the uploading and downloading efficiency of model parameters, and then affect the convergence speed and stability of the model. In some federated learning settings, the nodes involved in training are randomly selected, so the randomness of the faulty nodes can result in different nodes participating in each training, which can also lead to fluctuations in model performance. The problem of data loss also cannot be ignored. The faulty node may contain important training data, and its absence will lead to incomplete information during model training, thus affecting model performance. Node failures also reduce data diversity and reduce the generalization ability of the model, which in turn affects the performance of unseen data.

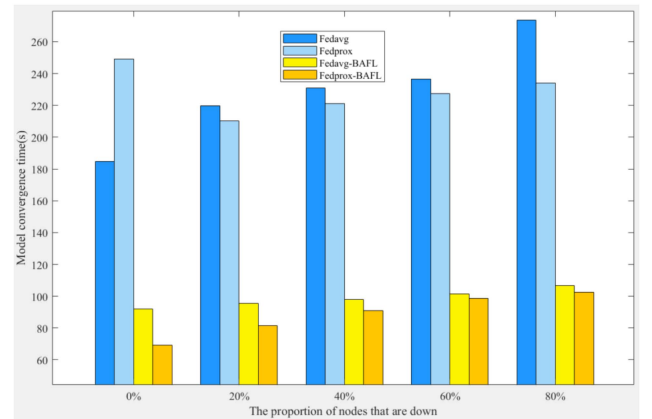


Fig. 11. Convergence time.

During the training process, Fedprox and Fedavg models showed obvious performance degradation after node failure, and slowly resumed training after node recovery. As can be seen from Tables V–VIII and Figs. 7–11, when acc converges, the number of iteration has reached 17 and 18, and the required time is 210.4 seconds and 219.6 seconds, respectively. Due to the introduction of a time attenuation factor in our model, which can effectively eliminate the influence of the old parameters of the long-failed nodes, the acc and loss of Fedprox-BAFL and Fedavg-BAFL only fluctuate slightly when the nodes fail, and the training will resume quickly after the nodes return to normal. Acc reached convergence in the 10th and 11th iteration, with times of 81.56 seconds and 95.56 seconds, respectively. In contrast, our proposed BAFL algorithm exhibits smaller fluctuations in the case of node failure and can reach convergence more quickly. It can be seen from the data in the above figure and

TABLE VIII
MODEL PERFORMANCE COMPARISON WHEN 80% OF CLIENTS ARE DOWN

	Precision	Recall	F1-score	Convergence time	LOSS	ACC
FEDAVG	0.96981	0.95544	0.96258	273.8s	0.2895	0.9622
FRDPROX	0.96984	0.95545	0.97712	234.1s	0.2789	0.9631
BAFL-FEDAVG	0.98031	0.98514	0.98272	106.7s	0.1804	0.9877
BAFL-FRDPROX	0.98058	0.99999	0.99019	102.5s	0.1608	0.9902

table that the convergence time and iteration times of the model increase correspondingly with the increase of the proportion of faulty nodes. From the perspective of model performance, the performance indexes of Fedprox-BAFL and Fedavg-BAFL, such as acc, loss, precision, recall and F1-score, did not decrease significantly in the case of node failure compared with the case without fault. Fedprox and Fedavg models show obvious performance degradation after node failure.

The temporal decay factor employed by the BAFL model demonstrates remarkable efficacy in mitigating the potential interference of old parameters from long-term offline nodes on current model training. Even if some clients remain offline for extended periods, their old parameters are unlikely to exert a significant negative impact on the ongoing model training process. For instance, when 20% of the nodes fail, the accuracy and loss value of the BAFL model only exhibit minor fluctuations, and once the faulty nodes return online, the model can swiftly resume its normal training trajectory. Moreover, the BAFL model relies on an asynchronous federated learning algorithm, which endows clients with the flexibility to update model parameters at different time points. This feature not only enhances the adaptability of the model training process but also effectively ensures the continuity and stability of model training in the face of client failures. Even if some clients are unable to update parameters in a timely manner due to failures, other clients can still continue to advance the model training, thereby ensuring the robustness of the model's overall performance.

C. Stable Performance of BAFL

In the previous part, we described in detail the experimental setup of collaborative participation in 20 clients. However, these experimental results are not sufficient to reveal the stable performance of BAFL (Blockchain-assisted Federated Learning) in larger alliances. To address this research gap, we introduced new clients to simulate the scenario where new business chains join the supervision chain and are allocated corresponding monitoring nodes and clients by the supervision chain. This part expands the assessment of BAFL performance to cover a range of growing league sizes from 5 to 80 clients, and quantifies key performance metrics, including F1-score, time per round of aggregation, and total time to convergence. Specifically, we randomly suspended the training of some clients to simulate mid-process client failures, and introduced new clients to simulate the scenario where new business chains join the supervision chain and are allocated corresponding monitoring nodes and clients by the supervision chain.

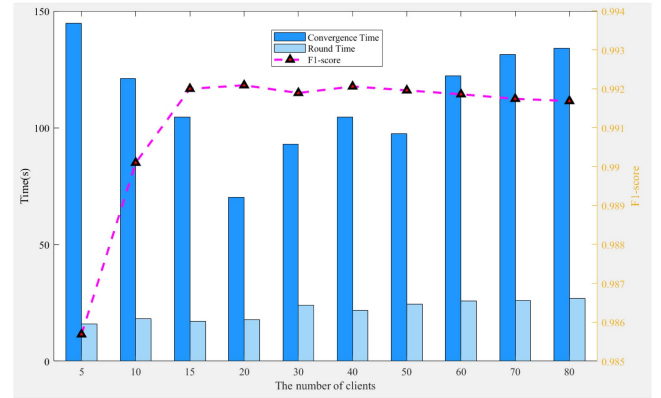


Fig. 12. The scalability of the Fedprox-BAFL.

For models with different number of clients, the experiment was repeated for 10 times, and the average trend of their F1 scores and convergence time was shown in Fig. 12. When the number of clients is small, the overall convergence time of the BAFL model is longer and the F1-score is relatively low. As the number of nodes increases, the model is able to process more data in parallel, which may speed up the training process. The addition of clients provides a more representative sample, which enhances the generalization of the model and improves the F1-score. When the number of clients increased to 20, the model showed the shortest convergence time. The different data perspectives provided by the new clients helped reduce the model's dependence on specific data biases, and enhanced the consistency and performance of the model on different data subsets, thus promoting the stability of F1-score. As the number of clients further increases, although the model's F1-score shows some stability, the network communication overhead increases due to the need for more communication to aggregate model updates. As the number of clients increases, so does the amount of data that needs to be transferred between the server and the client, which can lead to higher communication costs per training round, potentially slowing down training. In situations where bandwidth is limited or communication costs are high, the increase in communication overhead may further reduce the training speed and prolong the convergence time of the model, while the F1-score of the BAFL model fails to improve further.

Within the BAFL framework, the adoption of a DAG structured supervision chain enables efficient recording of model parameters and detected DDoS attack events. The DAG structure supports parallel confirmation of multiple transactions, thereby effectively circumventing the performance bottlenecks caused

by sequential processing in traditional blockchains and significantly enhancing system throughput. For instance, in the scenario where the number of clients increases, the DAG structure demonstrates higher processing efficiency, capable of rapidly handling and synchronizing model parameters, which in turn accelerates the model training process. Moreover, BAFL has automated data validation and sharing through smart contracts, ensuring the security and credibility of data. This mechanism allows newly joined clients to quickly synchronize the latest model parameters and seamlessly participate in model training. After joining, new clients can rapidly obtain the latest model parameters through the supervision chain, thereby substantially reducing initialization time and improving overall training efficiency.

VII. CONCLUSION

We summarize the research status and problems of DDoS attack detection in blockchain networks, and propose a new method to detect DDoS attacks in blockchain networks by integrating FL with the supervision chain. The proposed system, BAFL, leverages the decentralized and secure nature of blockchain technology to enhance the efficiency and robustness of DDoS attack detection while preserving the privacy of local node data. Through the use of a Directed Acyclic Graph (DAG)-structured supervision chain, BAFL ensures secure and transparent sharing of model parameters among distributed nodes. Additionally, the asynchronous federated learning algorithm, augmented with a time decay factor, optimizes the aggregation process by reducing the impact of outdated model parameters, thereby accelerating convergence and improving detection accuracy. The experimental results show that the accuracy of our proposed detection algorithm reaches more than 99%, showing excellent robustness and scalability. Compared to traditional DDoS attack detection algorithms, our solution can improve detection accuracy by 2% and reduce real-time detection latency to half of the industry's current methods.

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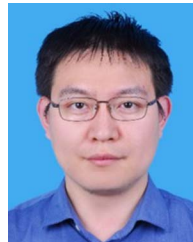
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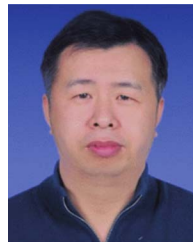


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